Igor Lucindo Cardoso

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Lubbock, TX

Objective

PhD student specializing in Operations Research, with a focus on large-scale optimization. Interested in integrating expertise in engineering and mathematical modeling to tackle complex real-world problems.

Education

Texas Tech University (TTU) — 2025 – Present
Ph.D. in Industrial Engineering — GPA: 4.00/4.00

- *Research interests:* Optimization, Mixed-Integer Programming (MIP), Graph Theory.

Getulio Vargas Foundation (FGV) — 2023 – 2024
M.A. Courses in Mathematical Modeling

- *Relevant coursework:* Measure Theory, Probability Theory.

Military Institute of Engineering (IME) — 2020 – 2024
B.S. in Electronic Engineering — GPA: 8.25/10.00

- IME is the oldest and one of the best-ranked engineering schools in Brazil.

Publications

- **Cardoso, I.**, Teodoro, G., Validi, H. (2026). **A Polytime and Interpretable Approach for Solving Wordle**. Submitted to *Operations Research*.
- Gutierrez, R., **Cardoso, I.**, Validi, H. (2026). **Sport Scheduling to Minimize Travels at the FIFA World Cup 2026**. Submitted to *Optimization Letters*.
- **Cardoso, I.**, et al. (2024). **Comparing Pre-Trained Object Detection Models for Autonomous Grasp on Affordable Prosthetic Hands**. *IEEE MeMeA*.
- *Manuscripts in Progress:* **Cardoso, I.** and Validi, H. **On Solving the Network-Based Portfolio Optimization Problems**. *In prep*, 2026.

Research Experience

Research Assistant, Texas Tech University (TTU) — 2025 – Present

- Conducting doctoral research in Operations Research, focusing on optimization, MIP, and graph theory.
- Developing novel algorithms for NP-hard combinatorial problems, leveraging GPU acceleration and decomposition techniques to solve large-scale instances. *Advisor: Dr. Hamidreza Validi.*

Research Experience for Undergraduates, Memorial University (MUN) — 2023

- Developed autonomous prosthetic grasping capabilities, optimizing machine learning and computer vision inference pipelines to achieve real-time predictive control. *Advisor: Dr. Vinicius P. da Fonseca.*

Honors & Awards

- Poster Finalist ($750 Travel Grant), MIP Workshop — 2026
- TTU Tye Industrial Endowment Engineering Scholarship, Texas Tech University — 2026
- Student Chapter Annual Award (Cum Laude), INFORMS — 2025
- 1st place in "The Road to 2050" competition, Shell Eco-marathon Team — 2021
- 5th place in Petrobras Drone Challenge, Latin American Robotics Competition — 2021
- Gold and Silver Medals, Brazilian Mathematical Olympiad of Public Schools (OBMEP) — 2018 – 2019

Extracurricular Experience

Ad-hoc Reviewer, *IISE Transactions* — 2026

- Conducted peer review for the flagship journal of the Institute of Industrial and Systems Engineers.

INFORMS Student Chapter, Texas Tech University — 2025 – Present

- Active member engaging with the research community in Operations Research.

Journal Club, Texas Tech University — 2025 – 2026

- Presenter in Journal Club sessions, analyzing peer-reviewed research in Operations Research.

American Airlines Scheduling Project — 2025

- Developed mixed-integer programming models to optimize aircraft scheduling for American Airlines.

Robotics Club - RoboIME — 2021 – 2023

- Contributed to projects involving machine learning and computer vision at the Military Institute of Engineering.

Software & Projects

Wordle Optimization Solver *Python, CuPy, CUDA*
Leveraged GPU-accelerated array operations for highly parallelized execution. [View App]

FIFA World Cup 2026 Scheduler *Python, JS, MIP Solvers*
Interactive web app using MIP to optimize match scheduling and minimize travel distance. [View App]

Expense Splitter *Python, Linear Programming*
Optimization tool to minimize the number of transactions required to settle shared expenses. [View App]

Computer Vision Rubik's Cube Solver *JS, OpenCV*
Built a real-time web app mapping physical states to a 3D digital twin to generate CFOP solutions. [View App]

Skills

Programming: C/C++, Python, Matlab, Julia, Java, JavaScript, HTML, CSS, FPGA, SQL
Tools: Gurobi, CUDA/GPU, Git, LaTeX, Linux, NetworkX, ROS, OpenCV, Arduino, Firebase
Languages: Portuguese (native), English (fluent), French (intermediate)
Coursework: Network Optimization, Measure Theory, Probability Theory, Machine Learning, Algorithms, Computer Vision, Computer Architecture.

References

Dr. Hamidreza Validi (advisor) - `hvalidi@ttu.edu`

Dr. Burak Eksioglu (professor) - `burak.eksioglu@ttu.edu`

Dr. Ningji Wei (professor) - `ningji.wei@ttu.edu`

MEMORIAL UNIVERSITY

Dr. Vinicius P. da Fonseca (advisor) - `vpradodafons@mun.ca`